A Two-Count Baseline for Change Detection on Dynamic Networks

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Code: github.com/ali-izhar/two-count-baseline

Abstract

Change-point detectors for dynamic networks aim to find changes in how a network is connected. They are evaluated on real networks with labelled events, but many of these events also change how active the network is: how many vertices take part and how many edges are present. A detector can therefore score well by responding to activity alone, without detecting any change in structure. We test whether detectors from the literature beat a simple baseline that tracks only the number of active vertices and the number of edges in each snapshot, a baseline that we motivate with a standard statistical test for a change in activity. On ten datasets, four of them from the papers of published detectors, none of thirteen detectors ranks the labelled events significantly better than the baseline when the results are pooled (AUC 0.75, or 0.73 with scales taken from early snapshots only, against 0.66 for the best detector). The baseline is best on only two datasets, however, and on the four datasets with the most events three detectors that track individual vertices score higher. When we rewire networks while keeping the degree of every vertex fixed, most detectors that ignore vertex identities barely respond, but several that track individual vertices do. New detectors should be compared with this baseline and tested at fixed degree.

1 Introduction

The MIT Social Evolution study recorded, day by day, which students of one dormitory were near one another (Madan et al., 2012). Each day gives a graph, or *snapshot*, whose edges join the students who were in contact, and the days form a series. Over Christmas 2008 the number of students active per day, those in at least one contact, fell from 34–43 to 3–8. A change-point detector scores each day by how much the recent snapshots differ from earlier ones, and it is judged on labelled events such as Christmas: it should score the days after the event above every stretch of the same length that contains no event, an *event-free window*. LAD (Huang et al., 2020), a detector built on the Laplacian spectrum, meets this standard at Christmas, and so does a score made of two counts: the number of active students and the number of pairs in contact. LAD's alarm is correct, but it cannot show that the pattern of contacts changed, because both counts changed with it.

Published detectors summarize each snapshot by a quantity that depends on its structure: the Laplacian eigenvalues (LAD), the distribution of the eigenvalues, or density of states (SCPD, Huang et al., 2023), or a similarity learned by a graph neural network (s-GNN, Sulem et al., 2024). *Permutation-invariant* detectors give the same score when the vertices are relabelled; *node-aligned* detectors compare the same vertices across snapshots. Either kind can respond to how many vertices and edges are present as well as to how they are connected.

Kendrick et al. (2018) found on weekly snapshots of two networks, MIT Reality Mining and Enron, that the number of active vertices and the number of edges, each fed to a Bayesian change-point method, detect the labelled changes nearly as well as summaries of fitted block and hierarchical models. The detectors published since (LAD, SCPD, s-GNN and MultiLAD, Huang et al., 2024) do not compare against these counts, and their benchmarks have further weaknesses. On AskUbuntu, whose events are Ubuntu releases six

months apart, a score that never changes reaches $F_1 = 1.00$ when each alarm is followed by a fixed quiet period: the alarms then recur at even intervals, as the events do. On MAG, and on two benchmarks that we cannot score, the labelled events are dates that the detector introducing the benchmark flagged and that its authors then matched to historical events. Last, we could not reproduce the reported results of s-GNN with its released code.

This paper asks what the benchmark events of these detectors change, and what the detectors see when activity is held fixed. We score thirteen detectors from the literature and two counts on the benchmarks of LAD, SCPD and s-GNN and on six other datasets, ranking each event against the event-free windows of its series.

1. A diagnosis of the benchmarks. Labelled events are large changes of activity far more often than event-free windows (an exploratory comparison), and pooled over ten datasets no detector ranks them significantly better than the two counts (Section 5.1).
2. A reason to use the counts as the control. A change of a graph splits into a change of activity and a change of structure, and in a Rasch model the window change of each count is the conditional score statistic for a change of activity (Section 3).
3. Tests at fixed degree, by rewiring real series and by block models of matched expected degree. Most permutation-invariant detectors barely respond to rewiring, while degree assortativity and several node-aligned detectors do (Section 5.2).
4. A protocol for evaluating a new detector of structural change (Section 6).

Compared with Kendrick et al. (2018), we evaluate the detectors that claim to see structure on their own benchmarks, rank events instead of thresholding scores, motivate the counts as the control, and test what the detectors see at fixed degree.

2 Related work

Activity as a confound. Many network statistics depend on the number of edges, so a change of that number can trigger or hide a detection (La Fond et al., 2014, 2018), as in brain networks (van Wijk et al., 2010). Peel and Clauset (2015) compared their generalized hierarchical random graph (GHRG) detector with mean-degree and clustering baselines; the size-agnostic detector of Miller and Mokryn (2020) remains sensitive to density by its authors' account, and surveys of graph distances report their dependence on size and density (Donnat and Holmes, 2018; Wills and Meyer, 2020; Hartle et al., 2020).

Trivial baselines and metrics. One line of code solves many time-series anomaly benchmarks (Wu and Keogh, 2023), and a method that never reports a change beats many change-point methods (van den Burg and Williams, 2020). In link prediction and graph classification, memorization and graph-free baselines compete with graph models (Poursafaei et al., 2022; Errica et al., 2020). Point-adjusted F_1 rates a random score highly (Kim et al., 2022), and range-based and volume-under-surface scores (Tatbul et al., 2018; Paparrizos et al., 2022) grade a whole series at once. We rank each event on its own and include a constant score.

Detectors we do not run. Model-based detectors fit a GHRG (Peel and Clauset, 2015) or monitor the block densities of a degree-corrected block model (Wilson et al., 2019), and a martingale detector tests the exchangeability of graph features (Ho et al., 2026); none is run on our datasets (Appendix A). Statistical work on correlation networks (Barnett and Onnela, 2016; Cribben and Yu, 2017), graphons (Zhao et al., 2019), missing links (Enikeeva and Klopp, 2025) and online detection (Yu et al., 2024) compares procedures with one another or with scan statistics.

3 Activity, structure and the counts

What a change can alter. A snapshot G_t is an undirected graph on a vertex set of size n, and a vertex is *active* if it has an edge. A change can alter four things: (i) the numbers of active vertices n_+ and of edges $|E|$; (ii) *membership*, which vertices are active; (iii) the *labelled structure*, which labelled pairs are connected, given (i) and (ii); and (iv) the *unlabelled structure*, the properties that do not depend on the vertex labels and are not fixed by the degrees, such as the triangles, the degree assortativity (Newman, 2002) or the blocks. *Activity* is (i) and (ii); *structure* is (iii) and (iv). Two students who swap all their contacts alter (iii) only. A family of scores can see a change if, for some graphs, the change alters at least one score of the family. A score built on n_+ and $|E|$ sees (i) only, and *vertex turnover*, the number of vertices that become active or inactive, sees (i) and (ii); neither can respond to structure, whatever the data. Table 1 shows which change each family can see, and where we test it.

Table 1. Which change each family of scores can see (a tick). Perm.-inv.: the permutation-invariant detectors and statistics such as transitivity and degree assortativity.

Change	Alters	Counts	Turnover	Perm.-inv.	Node-aligned	Tested in
A labelled event, e.g. Christmas at MIT	(i), (ii), possibly (iii), (iv)	✓	✓	✓	✓	Section 5.1
The same graph on other active vertices	(ii), (iii)		✓		✓	Section 5.1
Degree-preserving rewiring	(iii), (iv)			✓	✓	Section 5.2
Relabelling of the active vertices	(iii)				✓	Section 5.2

The counts. A window-based score compares a test window, the w snapshots before step t, with a reference window, the w_{ref} snapshots before those. We divide n_+ and $|E|$ by their scales, the median absolute change between consecutive snapshots, and write Δ_{n_+} and $\Delta_{|E|}$ for the changes of the window means of the two scaled numbers. The score at t is

$$s_t = \left((\Delta_{n_+})^2 + \lambda\,(\Delta_{|E|})^2\right)^{1/2},$$

with $\lambda = 1/4$, $w = 2$ and $w_{\text{ref}} = 5$, chosen as in Section 4; the weights $\lambda = 1/4$, $1/2$ and 1 give pooled EP within 0.002 of one another (Appendix A). We call this baseline score *the counts*.

A series may consist of *segments*, contiguous recordings such as one day of contacts or one subject's sensor data. Scales taken over the whole series use future snapshots, so the counts are offline; the *causal counts* take each scale from the first 20 snapshots of each segment only. We report both.

The counts and the score test. Suppose that a change of activity adds the same amount β to the log-odds that each unit (a vertex or a vertex pair) is present, while each unit keeps its own baseline rate. This is the Rasch model (Rasch, 1960) with a window effect: unit i is present in snapshot s with log-odds $\alpha_i + \beta\,\mathbf{1}[s \text{ in the test window}]$, independently across units and snapshots given the parameters. Let x_i and c_i count the test snapshots and all $N = w + w_{\text{ref}}$ snapshots in which unit i is present. Conditioning on c_i removes every α_i (Cox, 1958; Andersen, 1970), and the conditional score statistic for $\beta = 0$ is the Mantel–Haenszel numerator $\sum_i (x_i - w c_i/N)$ (Mantel and Haenszel, 1959). Counting the present units snapshot by snapshot gives, for any 0/1 presence array,

$$\sum_i \left(x_i - \frac{w\,c_i}{N}\right) = \frac{w\,w_{\text{ref}}}{N}\left(\bar{n}_{\text{test}} - \bar{n}_{\text{ref}}\right), \tag{1}$$

where $\bar{n}_{\text{test}}$ and $\bar{n}_{\text{ref}}$ are the mean numbers of present units per snapshot in the two windows. With vertices as units the right side is proportional to the window change of n_+; with pairs, to that of $|E|$. Each term of the counts is thus, up to a constant factor within a segment, the square of the score statistic for a change of activity at one level, with every unit's own rate conditioned away. The identity fixes the two terms but not their weights, so the counts are not the score test itself. Pooled, the standardized pair-level score test, which

divides by the conditional variance, ranks the labelled events of Section 5 as well as the counts, though not on every dataset; Appendix C gives the proof and these results.

4 Benchmark and protocol

Datasets. Table 2 lists the ten datasets. Four come from the papers of LAD, SCPD and s-GNN and are built with their loaders and events: UCI, MAG, the stablecoin network and PAMAP2, the *source datasets*; on MAG the events are SCPD's own detections. Three more datasets of these papers, the Senate, the Canadian bills and the flights, have all their events in the first 17 snapshots, which no method scores (Appendix B). We built MIT and Enron with our own event lists, MIT from the Social Evolution study with dates from the academic calendar (MIT Registrar's Office, 2008), and added four public datasets for their size and domain: the school, HT09, EEG and AskUbuntu. PAMAP2 and EEG are correlation networks, in which an edge joins two channels whose signals are correlated above a threshold; there $|E|$ measures how strongly the signals move together, and its change can be the event itself, as when a seizure starts. Every snapshot is a binary symmetric graph on a fixed vertex set; on AskUbuntu, each month's users are mapped to fixed user ids for the methods that need them.

Table 2. The datasets. T: snapshots; n: vertices of the fixed vertex set (AskUbuntu: users per month, at most); Events: labelled, with the number scored after the burn-in in parentheses; *: the source detector's own detections; Event-free: event-free windows; τ: tolerance in steps. Sources: LAD (Huang et al., 2020), SCPD (Huang et al., 2023), s-GNN (Sulem et al., 2024); ours: our construction and event list; added: a public dataset that we added.

Dataset	Source	Snapshot, events	T	n	Events	Event-free	τ
UCI messages	LAD	day, term ends and starts	196	1,900	2 (2)	177	0
MAG history	SCPD	year, wars, a regime change	181	502,549	5* (5)	139	2
Stablecoin	SCPD	day, market news	215	6,715,034	31 (30)	85	3
PAMAP2	s-GNN	1 s epoch, activity changes	19,261	27	98 (97)	18,016	5
MIT Social Evolution	ours	day, academic calendar	270	63	6 (6)	189	5
Enron, all addresses	ours	week, corporate events	200	86,664	4 (4)	147	4
Primary school	added	5 min, lunch	189	242	4 (4)	131	2
HT09 conference	added	5 min, sessions and breaks	434	113	22 (22)	283	2
EEG (CHB-MIT)	added	2 s epoch, seizure onsets	33,562	21	19 (19)	32,935	5
AskUbuntu	added	month, releases	76	$\leq$ 8,102	11 (9)	10	2

Event percentile. The constant score on AskUbuntu shows that F_1 at a tuned threshold rewards the alarm schedule as much as the score, so we score the ranking. Every method is scored from the same step, after a *burn-in* of the first 17 snapshots of each segment, the longest span of windows any method needs (LAD compares windows of 7 and 10 snapshots). An event at step e, its first changed snapshot, has the window $[e, e + \tau]$, where τ is the tolerance of its dataset; an event-free window is a run $[s, s + \tau]$ of scored steps within one segment that overlaps no event's window. The *event percentile* (EP) of an event is the fraction of event-free windows of its series in which a method's highest score is lower than in the event's window, with ties counted as one half: the AUC of the event window against the event-free windows. The constant score has EP 0.5. We average EP over the events of a dataset and then over datasets with equal weight, the *pooled EP*, so that PAMAP2's 97 events do not decide the result.

Detectors. We compare the counts with thirteen detectors and three structural statistics, the sixteen methods. Six detectors are permutation-invariant: LAD, SCPD, and four general change-point detectors that we run on eight summaries of each snapshot, among them $|E|$ and the mean degree. The four are the autoencoder TIRE (De Ryck et al., 2021), which is trained on the whole series and is therefore offline, the cumulative-sum test CUSUM (Page, 1954), Bayesian online change-point detection (BOCPD, Adams and MacKay, 2007) and maximum mean discrepancy (MMD, Gretton et al., 2012). Seven detectors are

node-aligned, each adapted from its published procedure to compare a test window with a reference window: Hamming and weighted Jaccard distances, DeltaCon (Koutra et al., 2013), the Chen–Zhang statistic (Chen and Zhang, 2015), network binary segmentation (NBS, Wang et al., 2021), adjacency spectral embeddings (ASE, Sussman et al., 2012; Madrid Padilla et al., 2022) and a SpotLight-style sketch (Eswaran et al., 2018). The *structural statistics* are the window changes of transitivity, of degree assortativity and the larger of the two; the triangle count grows with the number of edges, so transitivity also responds to activity.

Reproduction. LAD's code reproduces its reported results. SCPD's code recovers its detections on MAG and the flights; on the stablecoin network six of its eight reported detections fall one day later and none on the reported day. s-GNN's code reaches a mean point-adjusted F_1 of 0.14 on PAMAP2, for which its paper gives no settings, against 0.78 reported, so we leave s-GNN out (Appendix B).

What was fixed when. The settings of the counts and of CUSUM, BOCPD, MMD, LAD and TIRE were chosen by the highest mean F_1 on MIT, Enron and AskUbuntu before any EP was computed, so the other seven datasets are held out from that choice (Appendix A). The tolerances were fixed before any EP was computed, and the two pools (all ten datasets, and the four with at least ten events) and the equal weight per dataset were fixed at the first evaluation; the school, HT09 and EEG data and their events were fixed in a written protocol before any score based on the counts was computed on them. The causal counts, the node-aligned detectors and the structural statistics were added after the counts' first results. Analyses chosen after their own results are exploratory and marked °. The analysis specifications released with the code record these choices and the reading rules used below.

Inference. Each comparison of the counts with a method is tested twice, both times with Holm's correction (Holm, 1979) over the sixteen methods: by an exact sign-flip test over the per-dataset differences, and by a paired two-stage bootstrap that resamples datasets, then events, and gives the intervals. A t test over the per-dataset differences is reported beside them. We invert the two dataset-level tests, sign-flip and t, with Bonferroni's correction to bound how far any method's pooled EP can exceed that of the counts, and quote the larger of the two bounds. On the four datasets with at least ten events no sign-flip test can give p below 0.125, so there the bootstrap decides. With ten datasets the comparison has little power: an unadjusted test detects a lead of 0.12 to 0.25 in pooled EP with 80% power, depending on the method (Appendix D).

5 Results

5.1 Labelled events

Labelled events are large changes of activity three times as often as event-free windows. Averaged over the ten datasets, 45% of the events are large or extreme changes of activity°, against 15% of the event-free windows. To classify an event, we rank the window change of each count at the event among its changes over the event-free windows of the series, and call the change *large* when the larger of the two percentiles lies between 90 and 99, *extreme* above 99, and *ordinary* otherwise; an event-free window is ranked the same way. The share among events is higher on nine of the ten datasets; on MAG one of five events is large, against 20% of the event-free windows (Appendix D). Pooled over the datasets, every method ranks the extreme events above the ordinary ones (Figure 1(b)), and within single datasets every method except SCPD does so on average°. On the extreme events the counts have the highest EP, partly by construction.

Pooled, no detector ranks the events significantly better than the counts. Pooled over the ten datasets the counts have EP 0.751 and the causal counts 0.728; the best detector, Hamming, has 0.664, close to turnover and transitivity (Table 3). After Holm's correction the counts are significantly above BOCPD by both tests and above MMD and SCPD by the bootstrap; no method is significantly above either version of the counts. The simultaneous bounds limit any method's lead to 0.13 over the counts and 0.18 over the

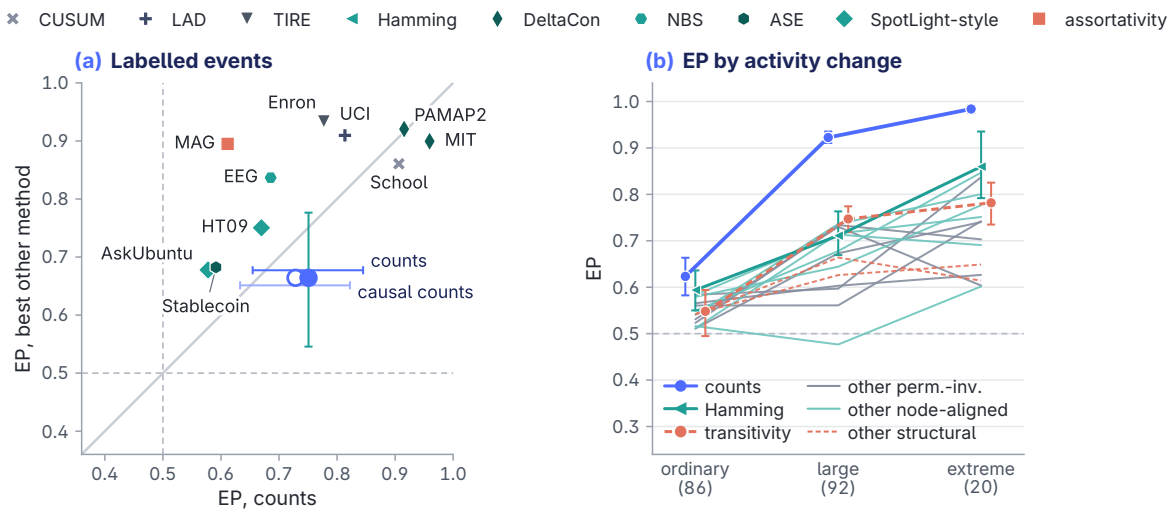


Figure 1. Labelled events. (a) EP of the counts (horizontal) and of the best of the sixteen methods on each dataset (vertical; the marker shows the method); choosing the best method after the fact favours it. Circles: pooled EP of the counts (filled) and of the causal counts (open) against that of Hamming, the best detector when pooled, with two-stage 95% intervals. (b) Pooled EP by the activity change at the event (Section 5.1); events in parentheses. The counts define these strata, so their rise is partly by construction.

causal counts. The counts are at or above every method only on MIT and the school network; on each of the other eight datasets some method is above them, but seven different methods take those places (Figure 1(a)). The counts keep the highest pooled EP under each check of Appendix D: without any one dataset, on the seven held-out datasets, without events taken from detectors' own detections, with the stablecoin events re-dated against the earliest public report°, and at common tolerances° from 1 to 10 (without AskUbuntu from $\tau = 3$ on). Only at a tolerance of zero, and with every event moved two steps earlier°, is another method ahead: TIRE, which is offline, and not significantly.

The lead is concentrated in events designed to change activity. We call an event *activity by design*° when its description names a scheduled break, holiday, meal or session, the start or end of a term, or a change in who takes part or in trading volume; the 37 such events are all those of MIT, the school, HT09 and UCI and three of the stablecoin network's. On them the counts have EP 0.827 and no detector or structural statistic has more than 0.695. On the 161 other events the counts have 0.690, and NBS (0.713), Hamming (0.708) and TIRE (0.694) are above them, none significantly.

On the datasets with the most events, three node-aligned detectors lead. On the four datasets with at least ten events, ASE, Hamming and NBS score higher than both versions of the counts. Their leads over the causal counts, 0.09 to 0.10, depend on the test and on the scale; SCPD, BOCPD, TIRE and the SpotLight-style sketch are also above the causal counts, by 0.02 to 0.04, none significantly. Pooled over the four datasets, only ASE's lead is significant, by the bootstrap, which decides there; it does not hold by the t test or with re-dated stablecoin labels (Appendix D). With each scale taken from all earlier snapshots of the segment, the *expanding scale*, the three leads fall to 0.01 to 0.03. ASE and Hamming also see membership (Table 1). Against vertex turnover, which sees only activity, the intervals of both leads cover zero, so by a reading rule fixed before turnover was scored the leads are not separable from turnover. Whether they see structure can only be tested with changes of structure at fixed activity.

5.2 Structural changes at fixed degree

Planted changes. To change structure at fixed activity, we replace 15 consecutive snapshots of a real series, from an onset on, by planted copies; a *plant* changes each copy and keeps n_+, $|E|$ and the degree

Table 3. Pooled EP, equal weight per dataset (datasets; events). Activity by design°: events whose description names a change of activity; Other: the remaining 161 events; ≥ 10 events: HT09, stablecoin, PAMAP2 and EEG. Counts − method: paired two-stage 95% interval, unadjusted; positive: the counts are above. After Holm's correction over the sixteen methods, †: significant by both tests; ‡: by the bootstrap only. Rows: references, permutation-invariant detectors, node-aligned detectors, structural statistics, each by descending EP on all ten. Bold: the highest detector or structural statistic per EP column. Differences from the causal counts on ≥ 10 events: Table 7.

	All ten datasets		Activity by design°	Other events	≥ 10 events
Method	EP	Counts − method	(5; 37 events)	(6; 161)	(4; 168)
Counts	0.751	–	0.827	0.690	0.716
Causal counts	0.728	+0.022 [−0.016, 0.068]	0.795	0.680	0.645
Constant	0.500	+0.251 [0.155, 0.345]	0.500	0.500	0.500
TIRE (offline)	0.645	+0.106 [−0.021, 0.246]	0.552	0.694	0.671
LAD	0.610	+0.141 [0.009, 0.271]	0.657	0.591	0.566
MMD	0.604	+0.147 [0.064, 0.227]‡	**0.695**	0.521	0.587
BOCPD	0.587	+0.163 [0.045, 0.307]†	0.603	0.575	0.682
CUSUM	0.573	+0.178 [0.037, 0.339]	0.513	0.622	0.592
SCPD	0.562	+0.189 [0.069, 0.329]‡	0.566	0.571	0.683
Hamming	**0.664**	+0.087 [−0.025, 0.222]	0.623	0.708	0.743
NBS	0.656	+0.095 [−0.026, 0.226]	0.568	**0.713**	0.730
DeltaCon	0.651	+0.099 [−0.023, 0.237]	0.570	0.642	0.608
Jaccard	0.627	+0.124 [0.008, 0.243]	0.610	0.600	0.611
ASE	0.615	+0.136 [0.001, 0.296]	0.550	0.683	**0.749**
SpotLight-style	0.608	+0.142 [0.002, 0.304]	0.661	0.586	0.664
Chen–Zhang	0.511	+0.240 [0.068, 0.423]	0.439	0.581	0.638
Transitivity	0.661	+0.090 [0.003, 0.175]	0.605	0.685	0.635
Max. of the two	0.597	+0.154 [0.021, 0.274]	0.577	0.571	0.540
Assortativity	0.590	+0.161 [0.024, 0.285]	0.619	0.549	0.488
Turnover°	0.665	+0.086 [−0.002, 0.177]	0.706	0.639	0.679

sequence; the rewirings also keep the degree of every vertex. Three plants rewire by double-edge swaps, aiming to move half the edges: at random, towards pairs of similar degree (assortative), or so as to close triangles. The fourth plant, *relabelling*, exchanges the labels of disjoint pairs covering half the active vertices; it changes only the labelled structure, so no permutation-invariant score can respond to it. A *trial* is one onset and one random seed. A method's *shift* is the mean increase of its highest score over the first five planted steps, compared with the same steps of the original series, in standard deviations of its highest scores over event-free windows; following the analysis specification, a shift below 0.1 standard deviations is negligible. Intervals need long series, so the results below come from HT09 and PAMAP2. On HT09 the triangle-closing plant moves almost no edge, which leaves five rewirings, three on PAMAP2 and two on HT09; Appendix E gives the school network and MIT.

Permutation-invariant detectors barely respond. Under the five rewirings, no permutation-invariant detector shifts upward by 0.1 standard deviations; the largest upward shift is TIRE's, 0.07 (Figure 2(a)). Some of these small shifts are consistent across trials but negligible. The only larger shift is downward: TIRE's score drops by 0.10 under HT09's assortative rewiring (Appendix E). Degree assortativity, a permutation-invariant statistic built for one of these changes, shifts by 0.2 to 2.0 standard deviations under four of the five. Relabelling leaves CUSUM, BOCPD, MMD, LAD and the structural statistics unchanged; SCPD changes only through its random probe vectors, and TIRE was not run on it.

Several node-aligned detectors respond, partly to lost persistence. Hamming shifts by more than 0.2 standard deviations under PAMAP2's random and assortative rewiring, and Jaccard by 0.45 under HT09's random rewiring; both also respond to relabelling. Each planted snapshot gets its own plant, so the planted series loses *edge persistence*, the tendency of the same pairs to be connected in consecutive snapshots, and part of the response is to this loss. A persistent plant, applied once to all fifteen snapshots, keeps persistence.

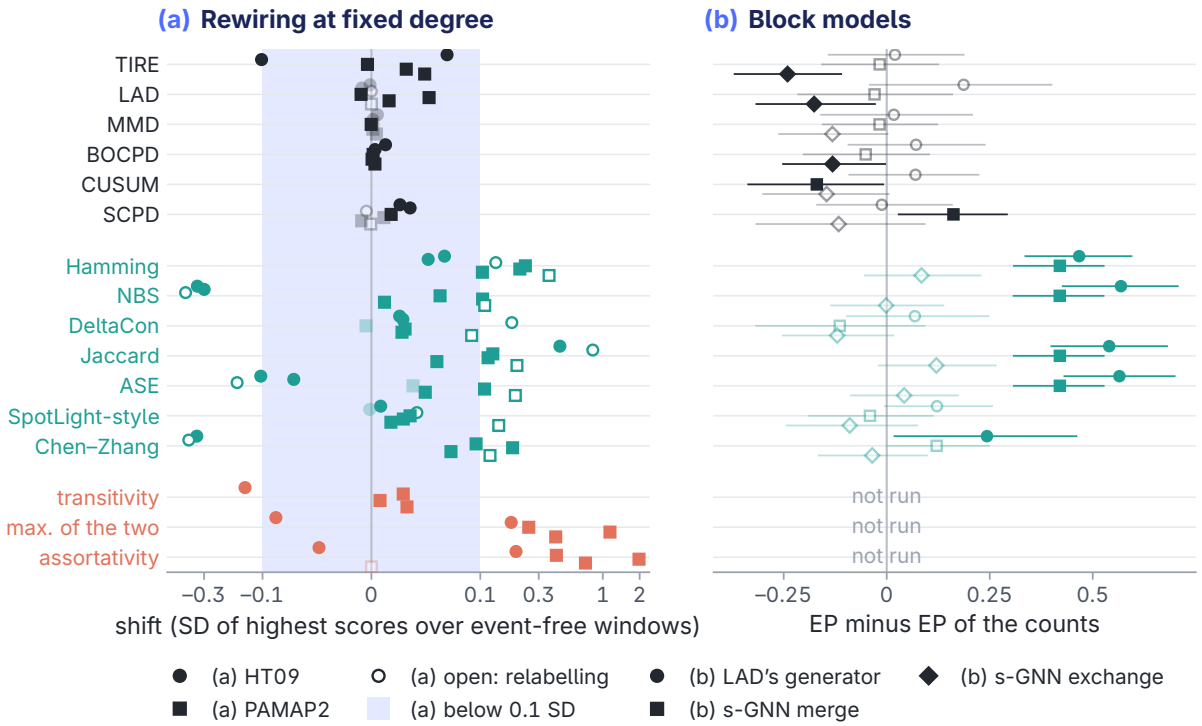


Figure 2. Structural changes at fixed degree. (a) Shift of each method's highest score after a plant on HT09 (circles) or PAMAP2 (squares), in standard deviations of its highest scores over event-free windows; filled: a rewiring; open: relabelling; faint: the 95% interval of the paired AUC, the share of trials in which the planted score exceeds the original score, covers one half. The axis is linear within ±0.1 (shaded) and logarithmic beyond. Left out: plants that move fewer than 1% of the edges, methods whose highest score is unchanged in at least half the trials (among them the counts), and TIRE on the relabelling, where it was not run. (b) EP minus the counts' EP, with 95% intervals, in the block models of LAD and s-GNN at matched expected degree and in s-GNN's vertex exchange; filled: the interval excludes zero. Colours as in Figure 1.

With it the responses of NBS, Hamming, Jaccard and DeltaCon to HT09's rewiring fall below 0.1 standard deviations; on PAMAP2 the persistent rewiring moves almost no edge and is not tested. Most responses to relabelling remain, among them Hamming's 0.50 standard deviations on PAMAP2 (Appendix E).

Block models at matched expected degree. We hold the expected degree fixed across each change in the generators of the LAD and s-GNN papers. In LAD's generator (1,000 vertices, mean degree 11), whose changes alter the number of blocks, the counts have EP 0.40, not distinguishable from the constant score, and no permutation-invariant detector, LAD included, differs significantly from the counts (EP 0.39 to 0.59), while NBS, Jaccard, Hamming and ASE reach EP 0.87 to 0.97 (Figure 2(b)). For local statistics this is expected: in sparse block models with equal expected degrees, the neighbourhood of a uniformly chosen vertex converges in distribution to the Poisson Galton–Watson tree of the Erdős–Rényi graph with that mean degree (van der Hofstad, 2024), in the sense of local weak convergence (Benjamini and Schramm, 2001; Aldous and Lyons, 2007). Vertex averages of bounded local statistics therefore have the same limit under all these models, and so do transitivity and degree assortativity under moment conditions on the degrees. Extreme eigenvalues, which LAD uses and which are among the summaries of CUSUM, BOCPD, MMD and TIRE, fall outside this argument. s-GNN's merge of four blocks into two is denser (mean degree 110) and outside the sparse limit; there SCPD, too, exceeds the counts by 0.16, as do four node-aligned detectors. In s-GNN's exchange of vertex pairs between blocks of equal density, a change of labelled structure only, no method is significantly above the counts. Without matched degree both changes also alter the degree, and the counts rank them at 0.97 and 1.00: such synthetic benchmarks reward a detector for seeing degree.

6 Discussion and conclusion

Implications for evaluation. On these benchmarks a detector's EP mixes its response to activity with whatever it sees of structure. The counts respond only to the numbers of active vertices and edges, a Rasch model motivates each of their terms as the score statistic for a change in one of these numbers, and in an exploratory comparison the events are large changes in them far more often than event-free windows. A detector that claims to see structure therefore has to beat the counts, offline and causal; pooled over the ten datasets none of the thirteen does so significantly, although on the four datasets with the most events three node-aligned detectors are ahead of the counts, and seven detectors of the causal counts. Degree assortativity and several node-aligned detectors respond to rewiring that the counts cannot see, the node-aligned detectors partly to lost persistence, while on the two long series the permutation-invariant detectors barely respond. The real benchmarks lack labelled events that change structure at fixed activity; until such events exist, tests at fixed degree are the direct evidence that a detector sees structure.

Limitations. Six datasets have fewer than ten events, and the events of one series share overlapping event-free windows but are resampled as independent, which can understate the variance. The labels are noisy, the tolerances follow no common rule, six of the ten datasets were built or added by us, and the settings of the counts and of five detectors were chosen on three datasets, MIT among them, which can favour settings that respond to activity. The node-aligned detectors are two-window adaptations, binary graphs can handicap detectors built for weights or directions, and we did not vary the correlation threshold of PAMAP2 and EEG. We ran no model-based detector on the ten datasets, no graph-kernel test and no graph neural network (we could not reproduce s-GNN's results).

A protocol for a new detector

(i) Report it beside the two counts, the causal counts and a constant score; report a node-aligned detector also beside vertex turnover.
(ii) Rank each event among the event-free windows of its series and pool with equal weight per dataset, instead of reporting F_1 at a tuned threshold.
(iii) Report the share of events whose activity change exceeds the 90th percentile of event-free changes, beside the same share of event-free windows.
(iv) Test at fixed degree, by rewiring with a relabelling control and persistent plants, and in block models at matched expected degree.
(v) Give a simultaneous bound on how far it could exceed the counts.

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A Detectors and settings

The exploratory methods and analyses (°) are turnover, fixed after the results on the four datasets with at least ten events; the split by event type, tagged after the main results but before any EP by type was computed, on datasets whose EP was known; and, chosen after their own results, the count variants, the equal weight per event, the activity strata and the share of event-free windows that reach them, the common tolerance, the label variants and shifts, the rules on event-free windows, the correlation-network subset and the Enron series of Peel and Clauset (2015). The reading rule for turnover (Appendix D) is in the released code and is not time-stamped. At $\tau = 0$ a score that sees only G_{t-1} and earlier cannot score an event's first changed snapshot.

Table 4. Scored methods (Section 4): the setting of each and when it was fixed, *before EP* (before any EP was computed), *after first results* (after the counts' first results against the permutation-invariant detectors) or *exploratory*. F_1: chosen by the highest mean F_1 over MIT, Enron and AskUbuntu. w, w_{ref}: test and reference windows; $w = 2$, $w_{\text{ref}} = 10$ for the node-aligned detectors except SpotLight-style and for the structural statistics. G_t: the score at t also sees G_t (otherwise G_{t-1} and earlier). Summaries, the eight inputs of TIRE, CUSUM, BOCPD and MMD: n, $|E|$, the mean and standard deviation of the degree over all n vertices, the density, the number of connected components, and the largest absolute adjacency eigenvalue twice (as spectral radius and as largest singular value, equal for a symmetric matrix).

Method	Setting	Fixed
Counts	scale: whole series; $w = 2$, $w_{\text{ref}} = 5$; 1/4 on the squared edge term (F_1)	before EP
Causal counts	scale: first 20 steps of each segment	after first results
Constant	the same value at every step	before EP
CUSUM (G_t)	two-sided on the summaries; threshold 45, $\kappa = 1.5$, start-up 10, cooldown 7 (F_1)	before EP
BOCPD (G_t); MMD	on the summaries; hazard 0.1; window 3 (F_1)	before EP
LAD (G_t); SCPD (G_t)	5 eigenvalues (F_1), windows 7, 10; 20 moments, its paper's windows 5, 10	before EP
TIRE (offline)	on the summaries; window 5, time and frequency domains, 100 epochs (F_1)	before EP
Hamming, Jaccard, NBS	ℓ_1, weighted Jaccard, CUSUM at the split; window-mean adjacency matrices	after first results
Chen–Zhang	edge-count statistic at the split on the Hamming minimum spanning tree	after first results
DeltaCon; ASE	$g = 10$ vertex groups; distance of rank-4 spectral embedding estimates	after first results
SpotLight-style	k-NN distance of the sketch of G_{t-1} to the 10 before; 50 random subset pairs	after first results
Structural statistics	scale: first 20 steps of a segment	after first results
Turnover	ℓ_1 change of the 0/1 active-vertex vector, over n; $w = 2$, $w_{\text{ref}} = 5$	exploratory
Variants	edge count alone, also causal; expanding scale; counts at $w_{\text{ref}} = 10$; LAD with the 6 or 499 eigenvalues of its code; NetSimile	exploratory

Grids and selection. EP sees only the settings that change the score. Of the counts' 5,040 settings only 48 do (16 pairs of window lengths times three ways of combining the counts); the others set thresholds and cooldowns, which EP ignores. The other tuned methods have 2,736 such settings (CUSUM), 360 (TIRE), 60 (LAD), 12 (BOCPD) and 7 (MMD). CUSUM's score resets after each alarm, so its threshold changes the score; without resets its pooled EP is 0.563, and the counts exceed it by +0.188 [0.054, 0.336]. Weights 1, 1/2 and 1/4 on the squared edge term give pooled EP 0.751, 0.752 and 0.751; weight 0 leaves the active-vertex count alone and gives 0.683. The whole-series scale is one constant per count and series, so it changes the ranking only through the counts' relative weight. On the stablecoin network the first 20 steps, which set the causal scale, contain one scored event. At the windows of the node-aligned detectors ($w = 2$, $w_{\text{ref}} = 10$) the counts have pooled EP 0.738 and stay above every detector. The selection used three MIT dates that differ from the calendar dates of Appendix B; reselecting with the calendar dates changes only the thresholds of CUSUM and MMD and LAD's setting, and we keep the original selection.

Other rules for choosing settings. Among the permutation-invariant methods the counts keep the highest pooled EP under each of four other rules: the authors' defaults (0.744; the counts have no published default and take the node-aligned detectors' windows), the median setting of each grid (0.702), the best setting in hindsight (0.762) and, for each dataset, the setting best on the other nine (0.751; Hamming and NBS, tuned by the same rule over their windows, have 0.655 and 0.677). On the four datasets with at least ten events TIRE exceeds the counts when its setting is chosen in hindsight or on the other datasets.

LAD's eigenvalues. LAD keeps five Laplacian eigenvalues, chosen by F_1 from 5 to 20; its code uses 6 on UCI and the Senate, 338 on the Canadian bills and 499 on its synthetic data. With 6 its pooled EP is 0.609 (five: 0.610). With 499° it gains 0.085 [0.009, 0.161] on the four datasets with at least ten events but falls on UCI (0.91 to 0.68) and MAG (0.79 to 0.51), and stays 0.136 [0.042, 0.253] below the counts on all ten (pooled EP 0.615). The signature distance of NetSimile° (Berlingerio et al., 2012) has 0.651.

Not run. We do not run MultiLAD (Huang et al., 2024), the detectors of Idé and Kashima (2004), Miller and Mokryn (2020) and Wilson et al. (2019), embedding tests (Zambon et al., 2019), AnomRank (Yoon et al., 2019), the martingale detector of Ho et al. (2026) or edge-stream detectors such as MIDAS (Bhatia et al.,

2020), surveyed by Ranshous et al. (2015) and Akoglu et al. (2015). GHRG runs only on its authors' series (Appendix B).

B Data, events and per-dataset results

Construction. Every snapshot is binary and symmetric, without self-loops. Three datasets of the source papers are not scored, because their events fall in the burn-in: the US Senate (12 Congresses, 225 vertices, two events of collaboration extremes) and the Canadian bills (14 years, 734 vertices, two leadership changes) of LAD, and the COVID flights (30 weeks, 3,000 vertices, two travel bans) of SCPD; on the bills and the flights the events are the source detector's own detections. UCI, the Senate and the Canadian bills follow LAD's loaders (Opsahl and Panzarasa, 2009; Huang et al., 2020); MAG, the flights and the stablecoin transfers follow SCPD's (Huang et al., 2023). PAMAP2 (Reiss and Stricker, 2012) follows s-GNN's: per subject, one correlation graph per 1 s epoch of the 27 inertial channels, with an edge where $|r| > 0.2$. The school and HT09 data (Stehlé et al., 2011; Isella et al., 2011) are 5-minute contact graphs, one segment per day. EEG uses 2 s epochs of the bipolar channels of CHB-MIT cases (patients) 1, 3 and 5 (Shoeb, 2009), with an edge where $|r| \geq 0.5$. AskUbuntu has one answer graph per month, with Ubuntu releases as events (Miller and Mokryn, 2020); as the snapshots renumber each month's users, the node-aligned detectors and turnover map them to 159,316 user ids, and for AskUbuntu the order of construction and scoring is not recorded. The tolerances are ours except UCI's, zero, which is LAD's; the released protocol holds those of the school, HT09, stablecoin, PAMAP2 and EEG. With m event-free windows an event's EP moves in steps of $1/(2m)$.

MIT and Enron. MIT is the Social Evolution study (Madan et al., 2012): daily Bluetooth proximity among 63 students of one dormitory. We do not use the earlier Reality Mining study of the same lab, used by Peel and Clauset (2015) and Kendrick et al. (2018), because rebuilt it keeps only four event-free windows after the burn-in. We keep the days with more than five records and more than one edge: 270 days from 30 January 2008 to 24 June 2009, all but eight of them from October 2008 on. From then on the filter drops only 22 to 25 December and 3 January, so the Christmas event falls on 26 December. The six events follow the MIT calendar for 2008–09 (MIT Registrar's Office, 2008) and the federal holidays: Columbus Day (13 October 2008), Thanksgiving (27 November), Christmas, New Year, the spring vacation (23 March 2009) and the day after the last day of classes (15 May). Over Christmas the activity drop starts one step before the event. Enron has one weekly email graph on all addresses (Klimt and Yang, 2004) and six corporate events merged into four.

Source labels. SCPD's stablecoin labels are days of market news; we re-dated each against the earliest dated public report. Of its 38 labels (on 31 days), 21 move one day later, four two days later and two one day earlier, five are unchanged and six are removed, which leaves 25 scored events. With these dates the counts stay significantly above BOCPD, MMD, SCPD and Chen–Zhang by the bootstrap, and ASE's lead over the causal counts on the four datasets with at least ten events shrinks to +0.083 [0.004, 0.149] (Holm-adjusted $p = 0.48$). On UCI the edges drop from 162 to 12 three snapshots before the end-of-term label; we keep LAD's label, and Appendix D gives the pool without UCI.

Reproduction. LAD's code and construction give Hits@10 (the share of labelled events among the ten highest scores) 0.50 on UCI and Hits@2 1.00 on the Senate, as reported. For s-GNN we used the authors' code (DyNNet, commit 2cd2035) with identity features, window 20 and the default network, because the paper gives no settings for PAMAP2. Its point-adjusted F_1, which credits a whole labelled interval for one alarm in it, has mean 0.14 over 16 trainings. Tuned over the default and 12 configurations of the paper's grid for its S&P 500 data (104 trainings), it reaches 0.00 at the configuration chosen by validation and 0.20 at the best in hindsight, against 0.78 reported; the counts reach 0.84, because point adjustment inflates every score (Kim et al., 2022).

Table 5. EP of every method on each scored dataset (scored events in parentheses); rows as in Table 3. Bold: the highest detector or structural statistic per column (ties all bold). On PAMAP2 DeltaCon (0.921) is above the counts (0.916). Last two rows: share of large or extreme activity changes (Section 5.1) among the events and among the event-free windows, each window ranked like an event.

	UCI (2)	MAG (5)	Stablec. (30)	PAMAP2 (97)	MIT (6)	Enron (4)	School (4)	HT09 (22)	EEG (19)	AskU. (9)
Counts	.81	.61	.59	.92	.96	.78	.91	.67	.69	.58
Causal counts	.83	.68	.59	.79	.95	.78	.88	.52	.68	.58
TIRE	.30	.63	.46	.82	.87	**.93**	.53	.59	.81	.51
LAD	**.91**	.79	.50	.79	.76	.46	.44	.41	.56	.47
MMD	.88	.41	.43	.71	.78	.49	.70	.53	.68	.43
BOCPD	.61	.37	.53	.91	.89	.53	.30	.62	.67	.44
CUSUM	.30	.51	.61	.61	.28	.88	**.86**	.56	.59	.53
SCPD	.23	.43	.60	.87	.60	.72	.63	.72	.54	.27
Hamming	.48	.55	.65	.91	.45	.88	.78	.67	.75	.52
NBS	.43	.46	.55	.88	.64	.88	.56	.65	**.84**	.67
DeltaCon	.68	.72	.23	**.92**	**.90**	.91	.40	.69	.60	.48
Jaccard	.56	.79	.46	.76	.81	.32	.78	.52	.70	.56
ASE	.22	.68	**.68**	.91	.47	.68	.69	.66	.74	.41
SpotLight-style	.20	.65	.54	.74	.76	.31	.83	**.75**	.62	**.68**
Chen–Zhang	.13	.46	.44	.77	.34	.37	.61	.52	.82	.64
Transitivity	.50	.74	.52	.82	.77	.82	.73	.49	.70	.50
Max. of the two	.45	**.89**	.42	.76	.85	.35	.77	.48	.51	.49
Assortativity	.46	**.89**	.44	.61	.88	.40	.83	.43	.47	.49
Turnover°	.57	.44	.61	.75	.88	.87	.63	.72	.63	.54
Events°	.50	.20	.27	.78	.67	.50	.50	.32	.42	.33
Event-free°	.14	.20	.15	.10	.15	.14	.18	.12	.18	.10

Table 6. Each source detector (LAD's datasets, then SCPD's) and the counts on the detector's own construction (Own) and on our binary graphs (Binary), on the events both score (from step 7 on the Senate, bills and flights), so values can differ from Table 5. Rows with one or two events are descriptive.

			Own		Binary	
Dataset	Det.	Ev.	Det.	Counts	Det.	Counts
UCI	LAD	2	.497	.792	.908	.808
Senate	LAD	1	1.00	1.00	.500	1.00
Bills	LAD	2	1.00	.500	.300	.500
MAG	SCPD	5	.850	.625	.452	.625
Flights	SCPD	1	1.00	.063	1.00	.063
Stablecoin	SCPD	30	.394	.590	.601	.590

Per-dataset results. Table 5 gives every method's EP on each scored dataset; per dataset, nominal 95% intervals with two or four events covered only 50% and 77% of the time in a simulation, so single-dataset statements on UCI, MAG, Enron and the school network carry no weight. Table 6 asks whether our binary construction handicaps the source detectors: on their own constructions they lead the counts on three of the six series, the Canadian bills, MAG and the flights, and only MAG has more than two events. On our Enron the counts (0.777) trail six methods and turnover, among them TIRE (0.934) and CUSUM (0.875).

Peel and Clauset's series. GHRG is the one model-based detector we run, on the two series of Peel and Clauset (2015), which we rebuilt: MIT Reality Mining (Eagle and Pentland, 2006) and Enron with 151 addresses and 23 events (log weekly edge counts of our Enron and theirs correlate at 0.925). On their Enron at a tolerance of one week the counts have EP 0.563; CUSUM (0.710), SCPD (0.705), TIRE (0.610), GHRG's likelihood ratio (0.595) and LAD (0.576) are above them, and BOCPD, GHRG's significance level, MMD and the authors' baselines (0.306 to 0.481) below. Without GHRG, across both Enron constructions, event lists and tolerances of one and four weeks, CUSUM leads in seven of eight cases and the counts rank second to fourth of seven methods. Their Reality Mining series keeps four event-free windows, too few to report alone. Added to the ten datasets, the two series give the counts a pooled EP of 0.713 (TIRE next: 0.658).

C Proof of the score identity

The identity. Each present unit of a snapshot adds one to $\sum_i c_i$, and one to $\sum_i x_i$ if the snapshot is in the test window, so $\sum_i x_i = w\,\bar{n}_{\text{test}}$ and $\sum_i c_i = w\,\bar{n}_{\text{test}} + w_{\text{ref}}\,\bar{n}_{\text{ref}}$. For any 0/1 presence array, therefore, $\sum_i (x_i - w c_i/N) = w\,\bar{n}_{\text{test}}(1-w/N) - (w\,w_{\text{ref}}/N)\,\bar{n}_{\text{ref}} = (w\,w_{\text{ref}}/N)(\bar{n}_{\text{test}} - \bar{n}_{\text{ref}})$, because $1 - w/N = w_{\text{ref}}/N$.

The score statistic. Under the Rasch model, fix a unit i and condition on c_i. Each set of c_i snapshots that could contain i then has probability proportional to $e^{\beta x_i}$, where x_i counts the test snapshots in the set, and α_i cancels. The conditional likelihood is $\prod_i \binom{w}{x_i}\binom{w_{\mathrm{ref}}}{c_i - x_i} e^{\beta x_i}/Z_i(\beta)$, where $Z_i(\beta)$ sums the numerator over x_i. At $\beta = 0$, x_i is hypergeometric with mean $\partial_\beta \log Z_i(0) = w c_i/N$, so the score at $\beta = 0$ is $\sum_i (x_i - w c_i/N)$, the left side of (1).

The standardized test. The null law of x_i given c_i assumes that the snapshots are exchangeable for unit i; edge persistence or a trend in activity breaks this. The product likelihood above and the conditional variance of the score, the sum of the hypergeometric variances $\sum_i c_i(N - c_i)\, w\, w_{\mathrm{ref}}/(N^2(N-1))$, assume independent units; pairs that share a vertex need not be independent, and the score then remains the score of the independence likelihood, but this variance no longer holds. Standardized by this variance, the score test has pooled EP 0.751 with pairs as units, the same as the counts (counts minus test 0.000 [−0.051, 0.048]), 0.716 with vertices as units and 0.679 for the joint test, the two standardized scores weighted by their inverse covariance.

D Inference and robustness

Coverage and power. In populations of ten datasets drawn from the observed ones, the two-stage interval of counts minus TIRE covered in 91% of the replicates, a t interval in 94%, and an interval that resamples events only in 51%. With four datasets the two-stage and t intervals covered in 96%, an optimistic estimate; the released specification keeps the bootstrap's Holm marks on the four datasets when this coverage is at least 90%, and reports the t test beside them. An unadjusted test detects a lead with 80% power when the lead is 2.80 two-stage standard deviations; depending on the method, this is 0.12 to 0.25 in pooled EP on the ten datasets and 0.09 to 0.23 on the four.

Tests and bounds. Every method is scored on the same 10,000 bootstrap resamples, so the differences are paired. The counts' pooled EP has the interval [0.655, 0.845] and the causal counts' [0.633, 0.822]. The sign-flip test is exact if the per-dataset differences are symmetric about zero; with ten datasets its smallest p is 0.002, or 0.031 after Holm's correction over sixteen methods. By the bootstrap, BOCPD, MMD and SCPD are below the counts after Holm's correction; by the exact test only BOCPD is (MMD 0.088, SCPD 0.19). Against the causal counts only MMD is below, by the bootstrap (Holm-adjusted $p = 0.048$). BOCPD and MMD read $|E|$ among their summaries, so two of these contrasts compare two uses of the same count. The counts stay first when any one dataset is left out (pooled EP 0.728 to 0.770), as do the causal counts (from 0.704). By the exact sign-flip test with Bonferroni's correction, no method's lead over the counts exceeds 0.128; by a t bound, 0.130; by the bootstrap's maximum studentized difference, 0.072. Over the causal counts the three bounds are 0.145, 0.177 and 0.104; Section 5.1 quotes the larger dataset-level bound of each. On the four datasets with at least ten events the exact bound is infinite, the Bonferroni t bound 0.505 and the bootstrap bound 0.137.

The four datasets with at least ten events. On HT09, the stablecoin network, PAMAP2 and EEG, ASE, Hamming and NBS are above the counts with Holm-adjusted $p = 1.00$ by the bootstrap and by t tests (Table 7). ASE's lead over the causal counts is +0.104: by the bootstrap its interval is [0.045, 0.157] and its Holm-adjusted p 0.03; by the t test its interval is [0.051, 0.156] and its Holm-adjusted p 0.126. Hamming's lead is +0.098 (Holm-adjusted p 0.076 by the bootstrap, 0.259 by t); no sign-flip test on four datasets can give p below 0.125. ASE's result is fragile: with the audited stablecoin dates its Holm-adjusted p is 0.48, and with a stage that resamples PAMAP2's eight subjects and EEG's three patients 0.46; on the ten datasets that stage barely moves the intervals. Leaving one dataset out moves the significance between detectors: without the stablecoin network NBS's lead over the causal counts, +0.126 [0.070, 0.195], is significant after Holm's correction and ASE's and Hamming's are not (Holm-adjusted $p = 0.107$ each); without EEG, ASE's (+0.117 [0.063, 0.168]) and Hamming's (+0.107, $p = 0.045$) are, and NBS's is not. The causal scale is the

Table 7. Pooled EP beyond Table 3: the four datasets with at least ten events and equal weight per event, with paired two-stage 95% intervals. ≥ 10 events (HT09, stablecoin, PAMAP2, EEG): EP and differences from the counts and the causal counts (negative: the method is above); †: significant after Holm's correction by the two-stage bootstrap, whose marks the specification keeps on these datasets; by the t test only the larger of the two structural changes (counts). Equal weight per event (all ten; PAMAP2 then has half the weight). Rows as in Table 3, then the variants (°: as in Section 4), whose pooled EP with equal weight per dataset is in the Variants paragraph and Appendix A. Bold: the highest detector or structural statistic per column. Expanding: scaled by all earlier steps of the segment. LAD, 499 eigenvalues: LAD with the number its code uses.

	≥ 10 events			Equal weight per event
Method	EP	Counts − method	Causal counts − method	EP, all ten
Counts	0.716	–	−0.071 [−0.146, 0.000]	0.791
Causal counts	0.645	+0.071 [−0.000, 0.146]	–	0.714
Constant	0.500	+0.216 [0.099, 0.354]	+0.145 [0.033, 0.257]	0.500
Permutation-invariant detectors (TIRE, CUSUM, BOCPD, MMD on snapshot summaries)				
TIRE (offline)	0.671	+0.045 [−0.082, 0.149]	−0.026 [−0.133, 0.094]	0.714
LAD	0.566	+0.150 [0.068, 0.250]†	+0.079 [0.002, 0.166]	0.655
MMD	0.587	+0.129 [0.033, 0.208]	+0.058 [−0.031, 0.145]	0.623
BOCPD	0.682	+0.034 [−0.027, 0.101]	−0.037 [−0.140, 0.062]	0.737
CUSUM	0.592	+0.123 [0.003, 0.256]	+0.052 [−0.042, 0.152]	0.592
SCPD	0.683	+0.032 [−0.060, 0.126]	−0.039 [−0.168, 0.094]	0.721
Node-aligned detectors (two-window adaptations)				
Hamming	0.743	−0.027 [−0.095, 0.031]	−0.098 [−0.161, −0.033]	**0.778**
NBS	0.730	−0.015 [−0.115, 0.064]	−0.086 [−0.168, 0.010]	0.764
DeltaCon	0.608	+0.108 [−0.023, 0.289]	+0.037 [−0.154, 0.267]	0.719
Jaccard	0.611	+0.105 [0.009, 0.184]	+0.034 [−0.049, 0.122]	0.666
ASE	**0.749**	−0.033 [−0.102, 0.026]	−0.104 [−0.157, −0.045]†	0.774
SpotLight-style	0.664	+0.052 [−0.052, 0.154]	−0.019 [−0.165, 0.096]	0.684
Chen–Zhang	0.638	+0.078 [−0.071, 0.195]	+0.007 [−0.115, 0.128]	0.654
Structural statistics				
Transitivity	0.635	+0.080 [−0.004, 0.169]	+0.009 [−0.052, 0.078]	0.706
Max. of the two	0.540	+0.176 [0.114, 0.247]†	+0.105 [0.021, 0.197]	0.633
Assortativity	0.488	+0.228 [0.138, 0.310]†	+0.157 [0.083, 0.240]†	0.560
Turnover°	0.679	+0.037 [−0.052, 0.132]	−0.034 [−0.154, 0.065]	0.700
Causal counts, expanding°	0.719	−0.003 [−0.025, 0.022]	−0.074 [−0.146, −0.004]	0.800
Edge count°	0.710	+0.005 [−0.032, 0.042]	−0.066 [−0.132, −0.000]	0.790
Counts, $w_{\text{ref}} = 10$°	0.716	+0.000 [−0.041, 0.038]	−0.071 [−0.163, 0.016]	0.788
Causal edge count°	0.686	+0.029 [−0.012, 0.074]	−0.041 [−0.097, 0.024]	0.764
LAD, 499 eigenvalues°	0.650	+0.065 [−0.002, 0.148]	−0.006 [−0.108, 0.095]	0.736

reason: the first 20 steps of a segment are a poor reference on HT09, where each day starts with the morning arrivals, and on PAMAP2. Against the counts scaled by all earlier steps of the segment (the expanding scale°; pooled EP 0.757 on the ten and 0.719 on the four), ASE's and Hamming's leads fall to +0.030 [−0.047, 0.107] and +0.024 [−0.038, 0.086] (t intervals).

Vertex turnover. ASE and Hamming see membership as well as structure (Table 1), so after their leads on these four datasets we scored turnover, which sees membership but no structure. A reading rule written before it was scored says that their leads are not separable from turnover if the intervals of both minus turnover cover zero. Turnover has 0.679 there, and ASE and Hamming exceed it by +0.070 [−0.028, 0.165] and +0.064 [−0.029, 0.163], so by the rule their leads are not separable from turnover, though with four datasets the rule has little power. On PAMAP2 alone they exceed it by about 0.16, with intervals above zero; on HT09 turnover is above ASE by 0.061 [0.008, 0.115] (event-level interval).

Event types. An event is activity by design if any of its descriptions contains break, holiday, vacation, Christmas, New Year, Thanksgiving, Columbus, lunch, coffee, term, classes, session, workshop, keynote, volume (or vol) or "whales leave"; the three stablecoin events so tagged report record trading volumes or large holders leaving, so the stablecoin network has events of both types. With five and six datasets the

exact test cannot resolve the split (smallest unadjusted p 0.0625 and 0.031). By the bootstrap with Holm's correction the counts are significantly above eleven of the sixteen methods on the activity-by-design events and the causal counts above three; on the other events no method is significantly above either version, and the counts have the highest EP in 4% of the resamples. Counted as activity by design, AskUbuntu's releases let Hamming lead the counts on the other events (0.745 against 0.712). After the split was seen°: MIT, UCI and the school network add 0.097 to the counts' pooled lead over Hamming, which is 0.087 in all; on HT09, whose events are all activity by design, the counts do not lead; and the tag does not track the measured activity change (16 of the 37 activity-by-design events and 96 of the 161 others are large or extreme, 76 of the latter on PAMAP2). Table 8 gives the counts on these and other subsets.

Table 8. The counts on subsets of the events (upper block) and of the datasets (lower block), equal weight per dataset. Highest other method: the highest detector or structural statistic, chosen after the fact; counts minus that method with a two-stage 95% interval. Each block by descending counts minus method. Other variants: Table 9.

Datasets or events (datasets; events)	Counts	Causal counts	Highest other method	Counts − method
Activity-by-design events (5; 37)	0.827	0.795	MMD 0.695	+0.133 [−0.008, 0.302]
Other events (6; 161)	0.690	0.680	NBS 0.713	−0.024 [−0.122, 0.074]
Held out, other events (4; 148)	0.696	0.680	ASE 0.752	−0.056 [−0.127, 0.008]
Source datasets, other events (3; 129)	0.699	0.681	ASE 0.756	−0.057 [−0.135, 0.009]
Held out from the settings (7; 179)	0.742	0.711	Hamming 0.684	+0.058 [−0.033, 0.170]
Built or added by us (6; 64)	0.763	0.731	TIRE 0.707	+0.055 [−0.100, 0.224]
Source datasets (4; 134)	0.733	0.725	LAD 0.750	−0.017 [−0.190, 0.121]
Correlation networks (2; 116)	0.801	0.733	NBS 0.861	−0.060 [−0.209, 0.042]

Activity strata. An event's activity change is ranked among the event-free windows of its series, none of which overlaps it; each event-free window is ranked the same way, among the event-free windows that do not overlap it. For one count the share of event-free windows above the 90th percentile is then close to 10% by construction (10% for n_+, 11% for $|E|$, averaged over datasets), and for either count 15% (last rows of Table 5). The share of events is higher on nine datasets and slightly lower on MAG (0.200 against 0.201), exceeds one half only on MIT and PAMAP2, and is 41% on the eight datasets that are not correlation networks. Pooled over events it is 112 of 198, 76 of them PAMAP2's. On the 86 ordinary events the counts (0.623) and Hamming (0.594) do not differ significantly. Within each of the seven datasets with both ordinary and extreme events, EP on the extreme minus the ordinary events° is positive on average for every method except SCPD (−0.043); five of these datasets have one to three extreme events, so this contrast is descriptive.

Variants. Table 9 repeats the pooled comparison under each doubt about labels, event-free windows and alignment. With one tolerance on every dataset° (without AskUbuntu from $\tau = 3$ on, where it keeps no event-free window) both versions of the counts stay above every detector and structural statistic from $\tau = 1$ to 10 (counts 0.67 to 0.79), turnover is above the causal counts at $\tau = 9$ and 10, and TIRE leads at $\tau = 0$ (0.62 against 0.59); on the four datasets with at least ten events another method leads at every tolerance. Of the label shifts from two steps earlier to two later, the only one where another method leads is the shift two steps earlier. With equal weight per event (Table 7) the counts have 0.791, above Hamming (0.778), ASE (0.774), NBS (0.764), BOCPD (0.737), SCPD (0.721) and DeltaCon (0.719), while the causal counts (0.714) fall below these six and level with TIRE; none of the six is significantly above them by the two-stage bootstrap. On the two correlation networks° NBS is above the causal counts by 0.127 [0.070, 0.217], but with two datasets the dataset stage has almost nothing to resample. The count variants of Table 7 are within 0.023 of the counts on all ten datasets (expanding scale 0.757, edge count 0.742 [0.662, 0.820], $w_{\mathrm{ref}} = 10$ 0.738, causal edge count 0.728), so $|E|$ carries most of the result.

Table 9. The pooled comparison under variants beyond Table 8, each named by the concern it addresses: the counts, the highest detector or structural statistic and counts minus it (two-stage 95% interval). Ten datasets unless a row names nine or the four with at least ten events. Upper block: the counts lead; lower block: another method leads; every interval covers 0. The label variants and the rows on event-free windows are exploratory.

Concern: variant	Counts	Highest other method	Counts − method
Labels from detections: without MAG	0.766	NBS 0.678	+0.088 [−0.044, 0.234]
Label after the drop: without UCI	0.744	Hamming 0.685	+0.059 [−0.046, 0.199]
Stablecoin dates: audited	0.757	Hamming 0.668	+0.089 [−0.023, 0.225]
Stablecoin dates: without the stablecoin network	0.769	DeltaCon 0.699	+0.070 [−0.046, 0.214]
Mixed scales: event-free windows of the event's subject or case	0.752	Transitivity 0.676	+0.076 [−0.008, 0.160]
Event-free windows: none ≤ 17 steps after an event (nine datasets)	0.780	DeltaCon 0.677	+0.103 [−0.042, 0.270]
none $\leq w + w_{\text{ref}}$ of each method after an event (nine)	0.779	DeltaCon 0.678	+0.101 [−0.041, 0.270]
Alignment: every score at t uses G_t	0.787	NBS 0.698	+0.089 [−0.018, 0.197]
≥ 10 events, audited stablecoin dates	0.731	Hamming 0.752	−0.021 [−0.090, 0.037]
≥ 10 events without the stablecoin network	0.757	NBS 0.790	−0.033 [−0.150, 0.055]
Label dates: every event two steps earlier	0.634	TIRE 0.657	−0.023 [−0.124, 0.067]
Tolerance: $\tau = 0$	0.592	TIRE 0.624	−0.032 [−0.153, 0.121]

Residual design. An exploratory design that regressed each score on the window changes of n_+ and $|E|$ and ranked the residuals failed its calibration: the residuals of the constant score and of the counts, which should rank events at chance, have EP 0.589 [0.543, 0.632] and 0.615, so we report no residual EP.

E Planted and simulated structural changes

Plants. Every rewiring is a sequence of double-edge swaps, $(a, b), (c, d) \to (a, d), (c, b)$ on four distinct vertices whose new edges are absent; a swap keeps every degree. A plant accepts swaps until $\rho|E|/2$ are accepted, with $\rho = 0.5$, or stops after 50 times as many tries, so the rewirings move 14% to 40% of the edges and HT09's triangle-closing plant almost none. The random plant accepts every valid swap; the assortative plant only swaps with $d_a d_d + d_c d_b > d_a d_b + d_c d_d$, which raise degree assortativity at fixed degrees; the triangle-closing plant only swaps that add triangles. The relabelling draws, in snapshot t, round$(\rho\, n_+(t)/2)$ disjoint pairs of active vertices and exchanges the labels of each pair. The persistent variant draws one rewiring, or one relabelling of the vertices active at the onset, and applies it to all 15 snapshots. Onsets are 40 steps apart, with offsets 0 to 39 and ten seeds; a *cell* is one series with one plant. The *paired AUC* of a cell is the share of trials in which a method's highest planted score exceeds its highest original score at the same steps, ties counted as one half, with a 95% interval from a bootstrap over 40-step periods. A method tied in at least half the trials of a cell (the counts: all) is not evaluable there; cells with fewer than ten periods, those of the school network (four) and MIT (six), are descriptive, and HT09 has ten and PAMAP2 478. TIRE was run only on seeds 0 to 2 and offsets 0 and 20, and not on the relabelling.

Shift sizes. The released specification calls a response consistent but negligible when its paired interval lies above one half, its unpaired AUC is below 0.55 and its mean shift is below 0.1 standard deviations; Section 5.2 uses the last condition. In the five rewiring cells of HT09 and PAMAP2 that move edges, TIRE, CUSUM (counted by the released rule, although tied in at least half the trials) and SCPD shift consistently in three, BOCPD and LAD in two and MMD in one, each by less than 0.1 standard deviations; the only larger shift is TIRE's downward shift of 0.102 under HT09's assortative rewiring. Degree assortativity shifts by 0.196 under HT09's assortative rewiring and by 0.418, 1.976 and 0.722 under PAMAP2's three rewirings, and falls under HT09's random rewiring. Under HT09's random rewiring Jaccard shifts by 0.446, Hamming's upward shifts are negligible, and NBS and Chen–Zhang fall by about a third of a standard deviation. On PAMAP2 Hamming shifts by 0.233 and 0.210 under random and assortative rewiring. On the school network, too short for intervals, SCPD and TIRE shift by 0.39 and 0.56 under random rewiring, where the larger of the two structural changes shifts by 3.08.

Relabelling controls. On the relabelling cells the counts, CUSUM, BOCPD, MMD, LAD and the structural statistics give the planted and the original series the same score in every trial, to a relative difference below 10^{-9}; LAD's school interval [0.352, 0.479] excludes 0.5 only through this round-off. SCPD's school interval excludes 0.5 because SCPD draws its random probe vectors in vertex order, so a relabelling changes its draws; with probes redrawn on unchanged series, 4 of 40 intervals exclude 0.5.

Table 10. Paired AUC of node-aligned detectors, plants per snapshot (independent, the design of Section 5.2) / once for all 15 snapshots (persistent), $\rho = 0.5$; brackets: persistent 95% interval. Cells: series and plant (R random, A assortative, T triangle-closing, S relabelling). Moved: median over trials of the mean share of the original edges that the plant removes over the 15 planted snapshots, in the plants regenerated for the node-aligned detectors (the plants of Figure 2(a) differ on HT09). School: four periods, descriptive. Grey: not evaluable (tied in at least half the trials).

Cell	Moved	Hamming	NBS	ASE	Chen–Zhang
School R	.39/.22	.87 / .80 [.79, .83]	.19 / .79 [.78, .81]	.18 / .20 [.18, .24]	.25 / .58 [.46, .67]
School A	.39/.20	.86 / .78 [.76, .82]	.19 / .75 [.73, .75]	.21 / .22 [.18, .26]	.27 / .58 [.45, .67]
School T	.14/.07	.71 / .44 [.40, .52]	.28 / .69 [.67, .72]	.76 / .79 [.73, .82]	.46 / .56 [.51, .61]
School S	.72/.68	.95 / .95 [.93, .96]	.23 / .98 [.93, 1.00]	.20 / .81 [.75, .89]	.20 / .58 [.44, .72]
HT09 R	.40/.18	.79 / .64 [.61, .67]	.21 / .56 [.54, .58]	.28 / .44 [.41, .48]	.31 / .46 [.44, .48]
HT09 A	.19/.13	.72 / .60 [.56, .65]	.27 / .57 [.54, .60]	.37 / .55 [.49, .60]	.38 / .47 [.45, .49]
HT09 T	.05/.03	.64 / .52 [.49, .56]	.39 / .54 [.51, .56]	.49 / .58 [.52, .62]	.47 / .50 [.49, .50]
HT09 S	.68/.47	.83 / .79 [.76, .82]	.20 / .76 [.68, .82]	.26 / .54 [.49, .58]	.28 / .47 [.43, .51]

Persistence. With one plant, NBS rises above 0.5 in the school and HT09 cells that are not tied, Chen–Zhang moves to about 0.5, and Hamming stays above 0.5 on both relabellings, so its response is not only to lost persistence. Under HT09's persistent relabelling Hamming's shift falls to 0.096, just below 0.1, while NBS's (0.29) and Jaccard's (0.61) stay large. ASE stays near 0.2 under the school network's random rewiring, which lowers the sum of squares of the four largest eigenvalues that its score tracks in every trial; it rises clearly above 0.5 only on the persistent school relabelling (0.195 to 0.810), which keeps every eigenvalue. Under PAMAP2's persistent relabelling, which moves 32% of the edges, Hamming has paired AUC 0.87 [0.86, 0.88].

Block models. At matched degree the realized mean degree moves by at most 1.25% across a change. On its own generator LAD has EP 0.466 [0.337, 0.589] at its fixed setting (0.587 at matched degree) and 0.991 with the 499 eigenvalues of its code. At matched degree in LAD's generator the counts have EP 0.401; LAD exceeds them by +0.187 [−0.043, 0.402], Chen–Zhang by +0.24 [0.02, 0.46], and DeltaCon and the SpotLight-style sketch by +0.068 and +0.122 with intervals that cover zero. In s-GNN's merge at matched degree SCPD exceeds the counts by +0.162 [0.027, 0.294]. In s-GNN's exchange of 40 vertex pairs the counts have EP 0.599 [0.489, 0.705]; Hamming and Jaccard are 0.08 and 0.12 above them, not significantly.